Week 08 teacher guide

Negative numbers, coordinates, and inequalities

Keep this guide beside PowerPoint slides 439-503. Follow the numbered slide sequence, then use the matching answer section after the student has tried.

The seven days

Day 050. Saturday | pages 195-201 | slides 439-457
Negative numbers and distance from zero

Day 051. Sunday | pages 202-212 | slides 458-477
Coordinates and inequalities

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Weekly check

This week’s idea

A signed number gives a position relative to zero. Coordinates use two such positions. Inequalities describe all positions that meet a condition.

Use the built-in Booster review where it appears in the day’s work. It replaces ordinary practice. No separate Booster packet is needed.

PDF bookmarks jump to each lesson and answer section. The student workbook’s Sunday reference pages are for checking an example after an attempt. Keep this teacher guide and the answer-review slides out of view during independent work.

Saturday. Negative numbers and distance from zero

Day 050, pages 195-201 | PowerPoint slides 439-457

Goal: Order positive and negative numbers and describe their distances from zero.

Set out: number line from −10 to 10, counters, pencil.

Check first: Fractions and decimals on a number line.

Pacing: Retrieval 5; launch 5; explore 15; discuss 10; synthesis 5; exit 5; preview 5; slack 10. Total 60 min.

Explore: what to watch for

Locates negatives to the left of zero. Why it matters: Anchors order to direction. Tutor move: Ask which point is farther right.

Separates a number from its distance. Why it matters: Prevents order and absolute-value confusion. Tutor move: Ask for both the signed location and the unsigned distance.

Uses fractions or decimal equivalents accurately. Why it matters: Both name the same point. Tutor move: Accept either representation and check the unit scale.

Likely misconceptions

Orders negatives by their digits. Why: Extends positive-number rules unchanged. Tutor move: Walk rightward on a number line and read the labels.

Gives a negative absolute value. Why: Keeps direction in a distance measure. Tutor move: Count unit steps from zero.

Follow the slide numbers below. Let the student attempt the task before showing the model or the answer. Write on the matching workbook pages; use extra paper only when more room is helpful.

Slide 439. Week 8, Saturday

Topic: Negative numbers and distance from zero
Student: Watch and discuss

Say: Order positive and negative numbers and describe their distances from zero.

Ask: What do you already know about this idea?

Expect: Fractions and decimals on a number line.

If stuck: Ask which point is farther right.

Slide 440. Today

Topic: Negative numbers and distance from zero
Student: Watch and discuss

Say: We will leave time to explain and check.

Ask: What can a model help us see?

Expect: Numbers increase to the right. Absolute value measures distance from zero, regardless of direction.

If stuck: Read the learning goal together.

Slide 441. Recall | 3 min

Topic: Negative numbers and distance from zero
Student: Day 050 · Workbook p. 195 · R1, R2, R3, R4

Say: Work from memory. We will check together.

Ask: Which earlier model helps?

Expect: 1: 8.528; 2: x = 7; 3: 48 square cm; 4: $7.35

If stuck: Give time to draw before giving a hint.

Slide 442. Recall: answers | 2 min

Topic: Negative numbers and distance from zero
Student: Day 050 · Workbook p. 195 · R1, R2, R3, R4

Say: Check one answer at a time.

Ask: Which step would you change?

Expect: 8.528; x = 7; 48 square cm; $7.35

If stuck: Rebuild one missed item with its model.

Slide 443. Temperature | 5 min

Topic: Negative numbers and distance from zero
Student: Day 050 · Workbook p. 196 · P1

Say: Read P1. Give time to try before comparing methods.

Ask: P1: What does zero mean here? Which direction is positive?

Expect: P1: −4°C; +7 m; −$9. Zero means 0°C, sea level, or no money owed or held.

If stuck: P1: Name the reference point before placing the quantity on a number line.

Slide 444. Integers on a number line | 3 min

Topic: Negative numbers and distance from zero
Student: Day 050 · Workbook p. 196 · P2

Say: Read P2. Give time to try before comparing methods.

Ask: P2: Which number lies farther right? Is that the same as being farther from zero?

Expect: P2: Left to right: −6, −2, 1; least −6

If stuck: P2: Locate zero first. Count equal units left or right, keeping the unit scale fixed.

Slide 445. The model | 2 min

Topic: Negative numbers and distance from zero
Student: Watch and discuss

Say: Match each part of the model to a number in the problem.

Ask: What does one part represent?

Expect: Opposites have equal distances from zero.

If stuck: Ask which point is farther right.

Slide 446. Fraction and decimal negative numbers | 2 min

Topic: Negative numbers and distance from zero
Student: Day 050 · Workbook p. 196 · P3

Say: Read P3. Give time to try before comparing methods.

Ask: P3: Which number lies farther right? Is that the same as being farther from zero?

Expect: P3: -2.5, $-\frac{3}{4}$, 0.5

If stuck: P3: Locate zero first. Count equal units left or right, keeping the unit scale fixed.

Slide 447. Opposite | 2 min

Topic: Negative numbers and distance from zero
Student: Day 050 · Workbook p. 197 · P4, P5

Say: Read P4 and P5. Give time to try before comparing methods.

Ask: P4: Which point lies equally far from zero on the other side? P5: How far is the point from zero, regardless of direction?

Expect: P4: 9; -9; $-(-9) = 9$; P5: 8, 8, 0 units from zero

If stuck: P4: Fold or reflect the number line across zero. Keep the distance and change the direction. P5: Count unit steps from zero. A distance cannot be negative.

Slide 448. Number order versus absolute-value order | 2 min

Topic: Negative numbers and distance from zero
Student: Day 050 · Workbook p. 197 · P6

Say: Read P6. Give time to try before comparing methods.

Ask: P6: How far is the point from zero, regardless of direction?

Expect: P6: $-9 < -4$; $9 > 4$; -4°C is warmer

If stuck: P6: Count unit steps from zero. A distance cannot be negative.

Slide 449. Watch the thinking | 2 min

Topic: Negative numbers and distance from zero
Student: Watch and discuss · worked example

Say: Read the reason beside each step.

Ask: Where does the model appear in the calculation?

Expect: $-1.5 < -0.5$

If stuck: Return to the matching unit in the model.

Slide 450. Your turn | 2 min

Topic: Negative numbers and distance from zero
Student: Day 050 · Workbook p. 198 · F

Say: Complete the unfinished method.

Ask: Which step is still needed?

Expect: 7; 7; 7

If stuck: Compare with the previous example without copying its numbers.

Slide 451. Two ways to see it | 5 min

Topic: Negative numbers and distance from zero
Student: Watch and discuss · compare the same example

Say: Partner compare: show the same problem in two ways.

Ask: How does each part of the model match a step in the calculation? Partner compare: Which is greater, and which is farther from zero? Agree or disagree: Can two different numbers have the same absolute value? Show them.

Expect: Accept drawings and equations that preserve the same quantities and units. Separates a number from its distance.

If stuck: Ask for both the signed location and the unsigned distance.

Slide 452. A claim to check | 5 min

Topic: Negative numbers and distance from zero
Student: Day 050 · Workbook p. 198 · D

Say: Agree or disagree. Find the first step that fails.

Ask: Why does this mistake seem reasonable? How can a model check it?

Expect: $-10 < -3$ because -10 lies farther left.

If stuck: Find zero, then trace left to each point.

Slide 453. Words and meaning | 3 min

Topic: Negative numbers and distance from zero
Student: Watch and discuss

Say: Use each word while pointing to a model.

Ask: Which example shows the meaning most clearly?

Expect: -4 can mean four degrees below zero. -6 and 6 $|-6| = 6$

If stuck: Read one definition and draw its example.

Slide 454. In your own words | 2 min

Topic: Negative numbers and distance from zero
Student: Day 050 · Workbook p. 195 · S

Say: Use your own words and one example.

Ask: Partner compare: Which is greater, and which is farther from zero?

Expect: Numbers increase to the right. Absolute value measures distance from zero, regardless of direction.

If stuck: Point to the model and name what each part represents.

Slide 455. Exit ticket | 5 min

Topic: Negative numbers and distance from zero
Student: Day 050 · Workbook p. 199 · E

Say: Work independently. Include the units and the requested reasoning.

Ask: Read the problem once. What is it asking you to find?

Expect: -6, -2.5, $-\frac{1}{4}$, 1.5; number -6, absolute value 6

If stuck: Do not coach during the ticket. Use the sort after collecting it.

Slide 456. The week ahead | 5 min

Topic: Negative numbers and distance from zero
Student: Days 052-056 · Workbook pp. 213-222

Say: Each packet is a short practice session.

Ask: Which model might help you during the week?

Expect: A model connected to the current topics.

If stuck: Open the study guide to the matching example.

Slide 457. Tutor reference

Topic: Negative numbers and distance from zero
Student: Tutor only · hidden in slide show

Tutor only. Keep this slide hidden during the lesson. Use it after collecting the exit ticket.

Exit answer: -6, -2.5, $-\frac{1}{4}$, 1.5; number -6, absolute value 6

Got it: Orders all four correctly and names -6 as the number with absolute value 6.

Almost there: Orders correctly but confuses the name -6 with distance 6, or makes one placement slip.

Needs re-teaching: Orders negatives by positive digits or selects 1.5 because it is the greatest number.

Sunday. Coordinates and inequalities

Day 051, pages 202-212 | PowerPoint slides 458-477

Goal: Use coordinates to locate and reflect points, and graph sets of solutions.

Set out: four-quadrant grid, ruler, blank number line.

Check first: Signed number order and horizontal or vertical distance.

Pacing: Coordinate plane 35 min; inequalities 25 min. Use retrieval 5, launch 5, explore 12, discuss 10, synthesis 3, mixed review 10, exit 5, preview 5, slack 5. Total 60 min.

Explore: what to watch for

Moves horizontally before vertically. Why it matters: Keeps coordinate order fixed. Tutor move: Have the student trace each move with a finger.

Preserves distance from the reflection axis. Why it matters: Shows a mirror relationship. Tutor move: Count equal squares on each side of the axis.

Finds distance by counting or subtracting positions. Why it matters: Both methods measure the same gap. Tutor move: Accept both and compare across zero.

Likely misconceptions

Swaps the coordinates. Why: Reads vertical position first. Tutor move: Use horizontal then vertical every time.

Changes both signs for an axis reflection. Why: Confuses reflection with a half-turn. Tutor move: Fold along the named axis.

Follow the slide numbers below. Let the student attempt the task before showing the model or the answer. Write on the matching workbook pages; use extra paper only when more room is helpful.

Slide 458. Week 8, Sunday

Topic: Coordinates and inequalities
Student: Watch and discuss

Say: Use coordinates to locate and reflect points, and graph sets of solutions.

Ask: What do you already know about this idea?

Expect: Signed number order and horizontal or vertical distance.

If stuck: Have the student trace each move with a finger.

Slide 459. Today

Topic: Coordinates and inequalities
Student: Watch and discuss

Say: We will leave time to explain and check.

Ask: What can a model help us see?

Expect: A coordinate pair gives horizontal then vertical position. An inequality can describe many solutions.

If stuck: Read the learning goal together.

Slide 460. Recall | 3 min

Topic: Coordinates and inequalities
Student: Day 051 · Workbook p. 202 · R1, R2, R3, R4

Say: Work from memory. We will check together.

Ask: Which earlier model helps?

Expect: 1: 20; 2: x = 25; 3: 62 square cm; 4: 7.714

If stuck: Give time to draw before giving a hint.

Slide 461. Recall: answers | 2 min

Topic: Coordinates and inequalities
Student: Day 051 · Workbook p. 202 · R1, R2, R3, R4

Say: Check one answer at a time.

Ask: Which step would you change?

Expect: 20; x = 25; 62 square cm; 7.714

If stuck: Rebuild one missed item with its model.

Slide 462. Plot points in all four quadrants | 5 min

Topic: Coordinates and inequalities
Student: Day 051 · Workbook p. 203 · P1

Say: Read P1. Give time to try before comparing methods.

Ask: P1: What is the horizontal position before the vertical position?

Expect: P1: A I; B II; C III; D IV

If stuck: P1: Start at the origin. Trace the horizontal move, then the vertical move.

Slide 463. Read coordinates from a graph | 2 min

Topic: Coordinates and inequalities
Student: Day 051 · Workbook p. 203 · P2

Say: Read P2. Give time to try before comparing methods.

Ask: P2: What is the horizontal position before the vertical position?

Expect: P2: P(−2, 4); Q(3, −1)

If stuck: P2: Start at the origin. Trace the horizontal move, then the vertical move.

Slide 464. The model | 2 min

Topic: Coordinates and inequalities
Student: Watch and discuss

Say: Match each part of the model to a number in the problem.

Ask: What does one part represent?

Expect: The y-axis lies midway between A and A′.

If stuck: Have the student trace each move with a finger.

Slide 465. Reflect across each axis | 2 min

Topic: Coordinates and inequalities

Student: Day 051 · Workbook pp. 203, 204 · P3, P4

Say: Read P3 and P4. Give time to try before comparing methods.

Ask: P3: Which coordinate stays fixed along the reflection axis? P4: Which coordinate changes? What is the full gap between the endpoints?

Expect: P3: Across x-axis: (−2, −3). Across y-axis: (2, 3).; P4: 7 units; 7 units

If stuck: P3: Fold the grid along the named axis and count equal distances on opposite sides. P4: Count to zero and then beyond it if the segment crosses an axis. Add the two distances.

Slide 466. Polygon perimeter on coordinate plane | 2 min

Topic: Coordinates and inequalities

Student: Day 051 · Workbook p. 204 · P5

Say: Read P5. Give time to try before comparing methods.

Ask: P5: Which coordinate changes? What is the full gap between the endpoints?

Expect: P5: 16 units

If stuck: P5: Count to zero and then beyond it if the segment crosses an axis. Add the two distances.

Slide 467. Open and closed circles | 2 min

Topic: Coordinates and inequalities

Student: Day 051 · Workbook pp. 204, 205 · P6, P7

Say: Read P6 and P7. Give time to try before comparing methods.

Ask: P6: Does the boundary value satisfy the words in the problem?

Expect: P6: $x >$; 2: open circle, shade right, excludes 2. $x \geq$; 2: closed circle, shade right, includes 2.; P7: $h \geq 48$; yes

If stuck: P6: Test the boundary itself. Use a filled endpoint only when equality is allowed.

Slide 468. Watch the thinking | 1 min

Topic: Coordinates and inequalities

Student: Watch and discuss · worked example

Say: Read the reason beside each step.

Ask: Where does the model appear in the calculation?

Expect: 5 units

If stuck: Return to the matching unit in the model.

Slide 469. Your turn | 1 min

Topic: Coordinates and inequalities

Student: Day 051 · Workbook p. 205 · F

Say: Complete the unfinished method.

Ask: Which step is still needed?

Expect: (5, −2); 10 units

If stuck: Compare with the previous example without copying its numbers.

Slide 470. Two ways to see it | 5 min

Topic: Coordinates and inequalities

Student: Watch and discuss · compare the same example

Say: Partner compare: show the same problem in two ways.

Ask: How does each part of the model match a step in the calculation? Partner compare: Which coordinate stays fixed when reflecting across the y-axis? Agree or disagree: Does at least include the boundary? Test the boundary in the sentence.

Expect: Accept drawings and equations that preserve the same quantities and units. Preserves distance from the reflection axis.

If stuck: Count equal squares on each side of the axis.

Slide 471. A claim to check | 5 min

Topic: Coordinates and inequalities

Student: Day 051 · Workbook p. 205 · D

Say: Agree or disagree. Find the first step that fails.

Ask: Why does this mistake seem reasonable? How can a model check it?

Expect: Use $x \geq 6$, a closed circle at 6, and shade right.

If stuck: Test the exact boundary value in the words.

Slide 472. Words and meaning | 2 min

Topic: Coordinates and inequalities

Student: Watch and discuss

Say: Use each word while pointing to a model.

Ask: Which example shows the meaning most clearly?

Expect: (−2, 3) means left 2, up 3. Across the y-axis, (−2, 3) becomes (2, 3). $h \geq 10$ includes 10 and larger numbers. $x > 4$ includes 4.5 and 5, but not 4.

If stuck: Read one definition and draw its example.

Slide 473. Mixed review | 10 min

Topic: Coordinates and inequalities
Student: Day 051 · Workbook p. 206 · M1, M2, M3

Say: Choose a method for each problem.

Ask: What tells you which operation to use?

Expect: 4.528; 13; 16 square cm

If stuck: Name the quantities and units before calculating.

Slide 474. In your own words | 1 min

Topic: Coordinates and inequalities
Student: Day 051 · Workbook p. 202 · S

Say: Use your own words and one example.

Ask: Partner compare: Which coordinate stays fixed when reflecting across the y-axis?

Expect: A coordinate pair gives horizontal then vertical position. An inequality can describe many solutions.

If stuck: Point to the model and name what each part represents.

Slide 475. Exit ticket | 5 min

Topic: Coordinates and inequalities
Student: Day 051 · Workbook p. 207 · E

Say: Work independently. Include the units and the requested reasoning.

Ask: Read the problem once. What is it asking you to find?

Expect: A′(4, 1.5); distance 8 units

If stuck: Do not coach during the ticket. Use the sort after collecting it.

Slide 476. The week ahead | 5 min

Topic: Coordinates and inequalities
Student: Days 052-056 · Workbook pp. 213-222

Say: Each packet is a short practice session.

Ask: Which model might help you during the week?

Expect: A model connected to the current topics.

If stuck: Open the study guide to the matching example.

Slide 477. Tutor reference

Topic: Coordinates and inequalities

Student: Tutor only · hidden in slide show

Tutor only. Keep this slide hidden during the lesson. Use it after collecting the exit ticket.

Exit answer: A′(4, 1.5); distance 8 units

Got it: Gives (4, 1.5) and distance 8, with the y-coordinate fixed.

Almost there: Gives the correct image but makes one distance error, or states the correct rule with one sign slip.

Needs re-teaching: Reflects across the x-axis, changes both signs, or gives distance 4 because only one half is counted.

PowerPoint answer review

After Friday, outside the weekend lesson time. Say: Check one completed item at a time. Ask: Where does your method first differ from this one? Expect: The answers and worked reasoning printed in the complete keys below. If stuck: Rebuild the quantity with its model, then have the student repair the first incorrect step.

Keep these slides out of view during independent work. The index below follows the answer-review slides in order; every answer is also printed in the complete keys.

Slide 478. Week 8 answers

Next lesson: slide 504. Use this review section after Friday.

Slide 479. Negative numbers and distance from zero (1/4)

Items: S15:R1; S15:R2; S15:R3; S15:R4. Follow along: Day 050 · Workbook p. 195 · Items R1, R2, R3, R4

Slide 480. Negative numbers and distance from zero (2/4)

Items: S15:S; S15:P1; S15:P2; S15:P3. Follow along: Day 050 · Workbook pp. 195, 196 · Items S, P1, P2, P3

Slide 481. Negative numbers and distance from zero (3/4)

Items: S15:P4; S15:P5; S15:P6; S15:F; S15:D. Follow along: Day 050 · Workbook pp. 197, 198 · Items P4, P5, P6, F, D

Slide 482. Negative numbers and distance from zero (4/4)

Items: S15:E. Follow along: Day 050 · Workbook p. 199 · Items E

Slide 483. Coordinates and inequalities (1/5)

Items: S16:R1; S16:R2; S16:R3; S16:R4. Follow along: Day 051 · Workbook p. 202 · Items R1, R2, R3, R4

Slide 484. Coordinates and inequalities (2/5)

Items: S16:S; S16:P1; S16:P2; S16:P3. Follow along: Day 051 · Workbook pp. 202, 203 · Items S, P1, P2, P3

Slide 485. Coordinates and inequalities (3/5)

Items: S16:P4; S16:P5; S16:P6; S16:P7. Follow along: Day 051 · Workbook pp. 204, 205 · Items P4, P5, P6, P7

Slide 486. Coordinates and inequalities (4/5)

Items: S16:F; S16:D; S16:M1; S16:M2; S16:M3. Follow along: Day 051 · Workbook pp. 205, 206 · Items F, D, M1, M2, M3

Slide 487. Coordinates and inequalities (5/5)

Items: S16:E. Follow along: Day 051 · Workbook p. 207 · Items E

Slide 488. Study guide: Try it

Items: W08_Study_Try:1; W08_Study_Try:2; W08_Study_Try:3; W08_Study_Try:4; W08_Study_Try:5. Follow along: Day 051 · Workbook p. 211 · Items 1, 2, 3, 4, 5

Slide 489. Daily practice (1/3)

Items: Day052_Practice:1; Day052_Practice:2; Day052_Practice:3; Day052_Practice:4; Day052_Practice:5. Follow along: Day 052 · Workbook p. 213 · Items 1, 2, 3, 4, 5

Slide 490. Daily practice (2/3)

Items: Day052_Practice:6; Day052_Practice:7; Day052_Practice:8; Day052_Practice:9; Day052_Practice:10. Follow along: Day 052 · Workbook pp. 213, 214 · Items 6, 7, 8, 9, 10

Slide 491. Daily practice (3/3)

Items: Day052_Practice:11; Day052_Practice:12; Day052_Practice:13. Follow along: Day 052 · Workbook p. 214 · Items 11, 12, 13

Slide 492. Daily practice (1/3)

Items: Day053_Practice:1; Day053_Practice:2; Day053_Practice:3; Day053_Practice:4; Day053_Practice:5. Follow along: Day 053 · Workbook p. 215 · Items 1, 2, 3, 4, 5

Slide 493. Daily practice (2/3)

Items: Day053_Practice:6; Day053_Practice:7; Day053_Practice:8; Day053_Practice:9; Day053_Practice:10. Follow along: Day 053 · Workbook pp. 215, 216 · Items 6, 7, 8, 9, 10

Slide 494. Daily practice (3/3)

Items: Day053_Practice:11; Day053_Practice:12; Day053_Practice:13. Follow along: Day 053 · Workbook p. 216 · Items 11, 12, 13

Slide 495. Daily practice (1/3)

Items: Day054_Practice:1; Day054_Practice:2; Day054_Practice:3; Day054_Practice:4; Day054_Practice:5. Follow along: Day 054 · Workbook p. 217 · Items 1, 2, 3, 4, 5

Slide 496. Daily practice (2/3)

Items: Day054_Practice:6; Day054_Practice:7; Day054_Practice:8; Day054_Practice:9; Day054_Practice:10. Follow along: Day 054 · Workbook pp. 217, 218 · Items 6, 7, 8, 9, 10

Slide 497. Daily practice (3/3)

Items: Day054_Practice:11; Day054_Practice:12; Day054_Practice:13. Follow along: Day 054 · Workbook p. 218 · Items 11, 12, 13

Slide 498. Daily practice (1/4)

Items: Day055_Practice:1; Day055_Practice:2; Day055_Practice:3; Day055_Practice:4; Day055_Practice:5. Follow along: Day 055 · Workbook p. 219 · Items 1, 2, 3, 4, 5

Slide 499. Daily practice (2/4)

Items: Day055_Practice:6; Day055_Practice:7; Day055_Practice:8; Day055_Practice:9; Day055_Practice:10. Follow along: Day 055 · Workbook pp. 219, 220 · Items 6, 7, 8, 9, 10

Slide 500. Daily practice (3/4)

Items: Day055_Practice:11; Day055_Practice:12; Day055_Practice:13; Day055_Practice:14; Day055_Practice:15. Follow along: Day 055 · Workbook p. 220 · Items 11, 12, 13, 14, 15

Slide 501. Daily practice (4/4)

Items: Day055_Practice:16. Follow along: Day 055 · Workbook p. 220 · Items 16

Slide 502. Weekly check (1/2)

Items: Day056_Practice:1; Day056_Practice:2; Day056_Practice:3; Day056_Practice:4; Day056_Practice:5. Follow along: Day 056 · Workbook p. 221 · Items 1, 2, 3, 4, 5

Slide 503. Weekly check (2/2)

Items: Day056_Practice:6; Day056_Practice:7; Day056_Practice:8; Day056_Practice:9; Day056_Practice:10. Follow along: Day 056 · Workbook p. 222 · Items 6, 7, 8, 9, 10

Weekday teaching and review

Use one day at a time. Aim for 15-20 minutes, or 15 minutes during a holiday week. Accuracy and a clear explanation matter more than finishing quickly.

Day 052. Monday | pages 213-214

Ask for a model from Saturday before checking an answer. Let the student choose a method. After checking, have the student correct the first step that changed the quantity.

Built-in Booster review: Number lines and coordinates: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 053. Tuesday | pages 215-216

Connect the practice to Sunday's model. Ask what each number counts and how a multiplication check, diagram, or estimate supports the answer.

Built-in Booster review: Number lines and coordinates: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 054. Wednesday | pages 217-218

Let the student choose the method for each mixed problem. For the puzzle, ask which conditions are already met. Accept any solution that meets every condition.

Built-in Booster review: Weekly puzzle: items 13. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 055. Thursday | pages 219-220

Ask the student to explain the planted error before repairing it. Optional Stretch comes after the required work and may be skipped without delaying the next day.

Built-in Booster review: Area, surface area and volume: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 056. Friday | pages 221-222

Give the weekly check independently, without coaching or a calculator. Score the required reasoning as well as the answer. Afterward, choose one error to discuss calmly.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

After Friday

On a 10-item check: 9-10 correct is strong; 7-8 is on track; 5-6 needs another review of the missed topic; 0-4 calls for 20 minutes of re-teaching at the next session. Record the skill and one observation. Check it again with a fresh item at least a week later.

Complete answers for the week

Match the Day number, item ID, and student page before checking. Questions stay in the student workbook; these keys give every answer and the needed reasoning. Equivalent correct forms and sound alternative methods are welcome. An explanation item needs a reason, model, or valid check as well as its conclusion.

Negative numbers and distance from zero

Day 050, student workbook pages 195-199

R1 [D-ADD] | student p. 195

Answer: 8.528

Reasoning: Line up decimal points. Add digits in the same place-value columns.

R2 [E-ADD] | student p. 195

Answer: x = 7

Reasoning: Subtract 11 from both sides.

R3 [G-PARA] | student p. 195

Answer: 48 square cm

Reasoning: Use base × perpendicular height: 6 × 8.

Watch for 60: Uses the slanted side.

R4 [D-MULT] | student p. 195

Answer: $7.35

Reasoning: 245 cents times three equals 735 cents.

S [EXPLAIN] | student p. 195

Answer: Answers vary. Numbers increase to the right. Absolute value measures distance from zero, regardless of direction.

Reasoning: Accept an accurate explanation supported by a matching example.

P1 [N-CONTEXT] | student p. 196

Answer: −4°C; +7 m; −$9. Zero means 0°C, sea level, or no money owed or held.

Reasoning: Use negative for below the reference point or a debt.

Watch for +4°C for below zero: Ignores the reference point.

P2 [N-ORDER] | student p. 196

Answer: Left to right: −6, −2, 1; least −6

Reasoning: The leftmost point is the least.

Watch for −2 is least: Compares only the positive digits.

P3 [N-ORDER] | student p. 196

Answer: -2.5, $-\frac{3}{4}$, 0.5

Reasoning: -2.5 lies farther left than $-\frac{3}{4}$. Both are below 0.5.

Watch for $-\frac{3}{4}$, -2.5, 0.5: Uses distance from zero as order.

P4 [N-OPP] | student p. 197

Answer: 9; -9; $-(-9) = 9$

Reasoning: Reflect across zero once for each opposite.

Watch for $-(-9) = -9$: Treats two negative signs as one.

P5 [N-ABS] | student p. 197

Answer: 8, 8, 0 units from zero

Reasoning: Both -8 and 8 lie eight units from zero.

Watch for $|-8| = -8$: Gives a direction instead of a distance.

P6 [N-ABS] | student p. 197

Answer: $-9 < -4$; $9 > 4$; -4°C is warmer

Reasoning: -4 is farther right, but -9 is farther from zero.

Watch for $-9 > -4$: Orders by absolute value.

F [N-ABS] | student p. 198

Answer: 7; 7; 7

Reasoning: Use the worked example in the slides to check each step.

D [N-ORDER] | student p. 198

Answer: $-10 < -3$ because -10 lies farther left.

Reasoning: Uses whole-number order without considering direction. Find zero, then trace left to each point.

E [N-ORDER] | student p. 199

Answer: -6, -2.5, $-\frac{1}{4}$, 1.5; number -6, absolute value 6

Reasoning: Place all four points, then compare their distances from zero.

Watch for 1.5 has the greatest absolute value: Confuses greatest number with greatest distance.

Coordinates and inequalities

Day 051, student workbook pages 202-207

R1 [D-DIV] | student p. 202

Answer: 20

Reasoning: Multiply both dividend and divisor by 100, then divide by 7.

Watch for 0.2: Changing only the divisor changes the quotient. Scale both quantities by 100.

R2 [E-MULT] | student p. 202

Answer: x = 25

Reasoning: 2 equal parts total 10. Find one part, then all 5.

R3 [G-SA] | student p. 202

Answer: 62 square cm

Reasoning: Two faces of each type: 2 × (6 + 10 + 15).

Watch for 30: Uses volume.

R4 [D-SUB] | student p. 202

Answer: 7.714

Reasoning: Write 10.000 and regroup through the zero columns.

S [EXPLAIN] | student p. 202

Answer: Answers vary. A coordinate pair gives horizontal then vertical position. An inequality can describe many solutions.

Reasoning: Accept an accurate explanation supported by a matching example.

P1 [N-COORD] | student p. 203

Answer: A I; B II; C III; D IV

Reasoning: Read x first, then y. Their signs identify the quadrant.

Watch for A at (3, 2): Swaps x and y.

P2 [N-COORD] | student p. 203

Answer: P(−2, 4); Q(3, −1)

Reasoning: P is left 2, up 4. Q is right 3, down 1.

Watch for P(4, −2): Reads vertical first.

P3 [N-REFLECT] | student p. 203

Answer: Across x-axis: (−2, −3). Across y-axis: (2, 3).

Reasoning: A reflection keeps distance from the mirror line and changes the coordinate across it.

Watch for Across x-axis: (2, 3): Reflects across the wrong axis.

P4 [N-DIST] | student p. 204

Answer: 7 units; 7 units

Reasoning: Only one coordinate changes. Count across zero or subtract the endpoints.

Watch for 1 unit: Subtracts magnitudes instead of positions.

P5 [N-DIST] | student p. 204

Answer: 16 units

Reasoning: Horizontal sides are 5. Vertical sides are 3. Add 5 + 3 + 5 + 3.

Watch for 15 square units: Finds area instead of perimeter.

P6 [E-INEQ] | student p. 204

Answer: $x > 2$: open circle, shade right, excludes 2. $x \geq 2$: closed circle, shade right, includes 2.

Reasoning: The equality bar decides whether the endpoint belongs.

Watch for Both include 2: Does not distinguish strict and inclusive comparisons.

P7 [E-INEQ] | student p. 205

Answer: $h \geq 48$; yes

Reasoning: At least means equal to or greater than.

Watch for $h > 48$: Excludes the allowed boundary value.

F [N-REFLECT] | student p. 205

Answer: $(5, -2)$; 10 units

Reasoning: Use the worked example in the slides to check each step.

D [E-INEQ] | student p. 205

Answer: Use $x \geq 6$, a closed circle at 6, and shade right.

Reasoning: Treats at least as more than. Test the exact boundary value in the words.

M1 [D-ADD] | student p. 206

Answer: 4.528

Reasoning: Line up decimal points. Add digits in the same place-value columns.

M2 [E-EVAL] | student p. 206

Answer: 13

Reasoning: $3 \times 4 + 1 = 13$.

M3 [G-AREA] | student p. 206

Answer: 16 square cm

Reasoning: Multiply the two perpendicular side lengths.

E [N-REFLECT] | student p. 207

Answer: $A'(4, 1.5)$; distance 8 units

Reasoning: The y-coordinate stays 1.5. Both points are 4 units from the y-axis, so the gap is 8.

Watch for $(-4, -1.5)$; distance 3: Reflects across the x-axis.

Study guide: Try it

Day 051, student workbook pages 208-211

1 [N-ORDER] | student p. 211

Answer: $-7, -1, 2$

Reasoning: Read the numbers from left to right on a number line.

2 [N-ORDER] | student p. 211

Answer: −9, −4, 0

Reasoning: Read the numbers from left to right on a number line.

3 [N-REFLECT] | student p. 211

Answer: (3, 4)

Reasoning: Keep the coordinate along the y-axis fixed; change the other sign.

4 [N-REFLECT] | student p. 211

Answer: (2, 5)

Reasoning: Keep the coordinate along the x-axis fixed; change the other sign.

5 [N-REFLECT] | student p. 211

Answer: (4, −2)

Reasoning: Keep the coordinate along the y-axis fixed; change the other sign.

Daily practice

Day 052, student workbook pages 213-214

1 [N-FACT] | student p. 213

Answer: 30

2 [N-FACT] | student p. 213

Answer: 70

3 [N-FACT] | student p. 213

Answer: 40

4 [N-FACT] | student p. 213

Answer: 90

5 [G-PARA] | student p. 213

Answer: 48 square cm

Reasoning: Use base × perpendicular height: 6 × 8.

Watch for 60: Uses the slanted side.

6 [P-PERCENT] | student p. 213

Answer: 75 %

Reasoning: Compare the green count with the total. Rename the fraction as a rate per hundred.

7 [N-FACT] | student p. 213

Answer: 24

8 [N-ABS] | student p. 213

Answer: 5 units from zero

Reasoning: Count the unit steps between the number and zero.

9 [N-ORDER] | student p. 214 | Built-in review: Number lines and coordinates

Answer: −9, −4, 0

Reasoning: Read the numbers from left to right on a number line.

10 [N-ABS] | student p. 214 | Built-in review: Number lines and coordinates

Answer: 12 units from zero

Reasoning: Count the unit steps between the number and zero.

11 [N-ABS] | student p. 214 | Built-in review: Number lines and coordinates

Answer: 0 units from zero

Reasoning: Count the unit steps between the number and zero.

12 [N-OPP] | student p. 214 | Built-in review: Number lines and coordinates

Answer: 6

Reasoning: Reflect the point across zero.

13 [N-ORDER] | student p. 214

Answer: −10 < −3 because −10 lies farther left.

Watch for −10 > −3: Uses whole-number order without considering direction.

Daily practice

Day 053, student workbook pages 215-216

1 [F-EQUIV] | student p. 215

Answer: 0.5; 50%

2 [F-EQUIV] | student p. 215

Answer: 0.25; 25%

3 [F-EQUIV] | student p. 215

Answer: 0.75; 75%

4 [F-EQUIV] | student p. 215

Answer: 0.125; 12.5%

5 [G-SA] | student p. 215

Answer: 62 square cm

Reasoning: Two faces of each type: 2 × (6 + 10 + 15).

Watch for 30: Uses volume.

6 [F-QUOT] | student p. 215

Answer: $\frac{5}{6}$ of a bar

Reasoning: Split each bar into sixths and give each child five sixths.

7 [N-FACT] | student p. 215

Answer: 12

8 [N-DIST] | student p. 215

Answer: 9 units

Reasoning: One coordinate stays fixed. Count the gap in the other coordinate.

9 [N-REFLECT] | student p. 216 | Built-in review: Number lines and coordinates

Answer: (3, 4)

Reasoning: Keep the coordinate along the y-axis fixed; change the other sign.

10 [N-DIST] | student p. 216 | Built-in review: Number lines and coordinates

Answer: 7 units

Reasoning: One coordinate stays fixed. Count the gap in the other coordinate.

11 [E-INEQ] | student p. 216 | Built-in review: Number lines and coordinates

Answer: $x \geq 9$; yes

Reasoning: Use a closed circle if equality is allowed; otherwise use an open circle.

12 [E-INEQ] | student p. 216 | Built-in review: Number lines and coordinates

Answer: $x < 5$; no

Reasoning: Use a closed circle if equality is allowed; otherwise use an open circle.

13 [E-INEQ] | student p. 216

Answer: Use $x \geq 6$, a closed circle at 6, and shade right.

Watch for $x > 6$: Treats at least as more than.

Daily practice

Day 054, student workbook pages 217-218

1 [N-SCALE] | student p. 217

Answer: 102

2 [N-SCALE] | student p. 217

Answer: 10

3 [N-SCALE] | student p. 217

Answer: 21

4 [N-SCALE] | student p. 217

Answer: 30

5 [N-ABS] | student p. 217

Answer: 0 units from zero

Reasoning: Count the unit steps between the number and zero.

6 [N-OPP] | student p. 217

Answer: −9

Reasoning: Reflect the point across zero.

7 [G-PARA] | student p. 217

Answer: 24 square cm

Reasoning: Use base × perpendicular height: 8 × 3.

Watch for 40: Uses the slanted side.

8 [G-COMP] | student p. 217

Answer: 37 square cm

Reasoning: Subtract the corner area from the full rectangle.

9 [F-SCALE] | student p. 218

Answer: Larger; 8

Reasoning: Four thirds takes one whole six plus one third of six.

10 [P-PART] | student p. 218

Answer: 6 stickers

Reasoning: Twenty-five percent is one fourth. Divide the whole by 4.

11 [N-GCF] | student p. 218

Answer: Common factors: 1, 2, 3, 6; GCF 6

Reasoning: List factors of both numbers and choose the greatest shared one.

12 [N-GCF] | student p. 218

Answer: Common factors: 1, 2, 4; GCF 4

Reasoning: List factors of both numbers and choose the greatest shared one.

13 [N-REFLECT] | student p. 218 | Built-in review: Weekly puzzle

Answer: Start (−3, 2); after y-axis (3, 2); after x-axis (3, −2)

Reasoning: The vertical axis measures horizontal distance. The horizontal axis measures vertical distance. Each reflection changes one sign.

Watch for (−2, 3): Swaps the distances to the axes.

Daily practice

Day 055, student workbook pages 219-220

1 [N-EST] | student p. 219

Answer: 1000

2 [N-EST] | student p. 219

Answer: 11

3 [N-EST] | student p. 219

Answer: About 100

4 [N-EST] | student p. 219

Answer: About 20

5 [N-ABS] | student p. 219

Answer: 0 units from zero

Reasoning: Count the unit steps between the number and zero.

6 [N-OPP] | student p. 219

Answer: -9

Reasoning: Reflect the point across zero.

7 [N-REFLECT] | student p. 219

Answer: (3, 4)

Reasoning: Keep the coordinate along the y-axis fixed; change the other sign.

8 [G-TRI] | student p. 219

Answer: 18 square cm

Reasoning: Half of 9×4 is 18.

Watch for 36: Omits the half.

9 [G-AREA] | student p. 220 | Built-in review: Area, surface area and volume

Answer: 21 square cm

Reasoning: Multiply the two perpendicular side lengths.

10 [G-COMP] | student p. 220 | Built-in review: Area, surface area and volume

Answer: 38 square cm

Reasoning: Subtract the corner area from the full rectangle.

11 [G-SA] | student p. 220 | Built-in review: Area, surface area and volume

Answer: 66 square cm

Reasoning: Two faces of each type: $2 \times (9 + 12 + 12)$.

Watch for 36: Uses volume.

12 [G-VOL] | student p. 220 | Built-in review: Area, surface area and volume

Answer: $\frac{2}{3}$ cubic meters

Reasoning: Multiply the three edge lengths. Track the three length units as cubic meters.

13 [E-INEQ] | student p. 220

Answer: $x \geq 9$; yes

Reasoning: Use a closed circle if equality is allowed; otherwise use an open circle.

14 [E-INEQ] | student p. 220

Answer: Use $x \geq 6$, a closed circle at 6, and shade right.

Watch for $x > 6$: Treats at least as more than.

15 [N-SIGNED] | student p. 220

Answer: 2

16 [E-EQ] | student p. 220

Answer: $x = 4$

Reasoning: Subtract 3 from both sides, then divide both sides by 2.

Weekly check

Day 056, student workbook pages 221-222

1 [N-ABS] | student p. 221

Answer: 0 units from zero

Reasoning: Count the unit steps between the number and zero.

2 [N-OPP] | student p. 221

Answer: -9

Reasoning: Reflect the point across zero.

3 [N-REFLECT] | student p. 221

Answer: (3, 4)

Reasoning: Keep the coordinate along the y-axis fixed; change the other sign.

4 [N-ABS] | student p. 221

Answer: 12 units from zero

Reasoning: Count the unit steps between the number and zero.

5 [E-MULT] | student p. 221

Answer: $x = 9$

Reasoning: Divide both sides by 6.

Watch for 48: Subtracts instead of dividing.

6 [E-MULT] | student p. 222

Answer: $x = 3.2$

Reasoning: Divide both sides by 1.5; then check the product.

7 [G-SA] | student p. 222

Answer: 40 square cm

Reasoning: Add the square base and four half-base-times-height triangles.

8 [F-SCALE] | student p. 222

Answer: Larger; 8

Reasoning: Four thirds takes one whole six plus one third of six.

9 [P-PART] | student p. 222

Answer: 6 stickers

Reasoning: Twenty-five percent is one fourth. Divide the whole by 4.

10 [N-ORDER] | student p. 222

Answer: $-10 < -3$ because -10 lies farther left.

Watch for $-10 > -3$: Uses whole-number order without considering direction.